

Assignment 2

Advanced Computer Architecture

Date of Submission: 11/03/2026

1. What are the different types of pipeline architectures? Explain instruction pipelining, arithmetic pipelining and superpipelining
2. What are the key differences between pipeline processing and sequential execution? Provide a comparative analysis.
3. Discuss the role of vector processing in pipeline architectures. How is it used in high-performance computing?
4. how does SMP (symmetric multiprocessing) differs from AMP (asymmetric multiprocessing)? Provide examples of each.
5. What factors influence the partitioning of a program in a parallel computing environment? Explain how load balancing is achieved through effective program partitioning. What strategies help in maintaining equal workload across processors?